

# Predictive Analytics in India's Ayushman Bharat PM-JAY Scheme: A Scalable Framework for Healthcare Fraud Detection

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## Abstract

This paper addresses the critical challenge of systemic fraud within India's Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (PM-JAY), the world's largest government-funded health insurance scheme. Utilizing large-scale healthcare transactional data and advanced machine learning frameworks, this study develops a predictive model to identify and mitigate fraudulent activities. By analyzing claim distributions across four states with varying degrees of digital infrastructure, the empirical results demonstrate that states with higher Electronic Health Record (EHR) adoption exhibit a significantly lower incidence of provider-level fraud. Furthermore, an end-to-end distributed data architecture is proposed to facilitate real-time anomaly detection, providing a proactive mechanism to safeguard public healthcare resources.

**Keywords:** Healthcare fraud detection, PM-JAY, big data analytics, machine learning, PySpark, state-wise comparison visualization, ABDM.

## 1. Introduction

### 1.1 Background

India's healthcare ecosystem serves an exceptionally large and diverse demographic of over 1.4 billion people. The system faces structural constraints, including disparities in clinical infrastructure, uneven hospital distribution, and a low doctor-to-patient ratio. To mitigate financial vulnerabilities and ensure universal health coverage, the Government of India launched the Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (PM-JAY) in 2018. Designed to provide financial protection to bottom-quintile socio-economic groups, the scheme covers secondary and tertiary hospitalization expenses. However, the scale and decentralized execution of PM-JAY make it susceptible to leakage, primarily driven by provider-side fraud, such as unbundled claims, ghost billing, phantom patients, and medically unnecessary treatments.

## 1.2 Statement of the Problem

The implementation of PM-JAY is highly decentralized, with State Health Agencies (SHAs) exercising operational autonomy. Consequently, substantial disparities exist in digital maturity, auditing rigor, and Electronic Health Record (EHR) integration across states. This fragmentation allows systematic anomalies and opportunistic fraud to go undetected in low-maturity states, resulting in capital flight that threatens the long-term fiscal sustainability of the scheme.

## 1.3 Research Aim and Objectives

This study aims to develop a robust, scalable, and automated predictive framework utilizing machine learning to minimize financial leakage in PM-JAY. The specific objectives are:

- To evaluate the relationship between state-level digital health maturity and fraud incidence.
- To design a multi-tiered machine learning pipeline optimized for highly imbalanced clinical data.
- To architect a distributed, real-time data engineering workflow capable of generating risk scores during the pre-authorization and claim adjudication cycles.

## 2. Review of Literature

### 2.1 Healthcare Fraud Typology in the Context

Healthcare fraud manifests through complex configurations of provider, patient, and intermediary collusion. Common typologies within public insurance contexts include upcoding (billing for more expensive procedures than performed), unbundling (billing service components separately to maximize payout), and phantom billing.

### 2.2 State-Wise Comparative Dimensions

Extant research underscores that the efficacy of anti-fraud strategies relies heavily on underlying provincial administrative capacity. States with legacy digital infrastructure often experience higher latencies in post-payment audits, making them vulnerable to sustained, low-velocity fraud rings.

### 2.3 Global Machine Learning Methods

Supervised learning models (e.g., Gradient Boosted Trees, Random Forests) are highly effective at detecting historically documented fraud patterns. However, they are fundamentally limited when encountering novel, evolving fraud strategies. Unsupervised anomaly detection frameworks (e.g., Isolation Forests) are crucial for identifying zero-day structural variations without relying on historical labels.

## 2.4 Big Data Architecture for Indian Healthcare

Given the high volume of daily transactions generated under PM-JAY, standard monolithic data warehousing systems fail to meet operational SLAs. Modern distributed computing frameworks, particularly Apache Spark (PySpark), offer the necessary horizontal scalability to handle massive parallel processing pipelines for near-real-time auditing.

## 2.5 Identified Research Gap

While literature extensively explores machine learning applications in Western insurance models (such as US Medicare), there is a distinct shortage of empirical research focused on large-scale, state-comparative, distributed predictive analytics tailored specifically to the structural nuances of the Indian public healthcare delivery system.

# 3. Theoretical Framework

## 3.1 Conceptual Model

This study synthesizes three foundational paradigms to map the behavioral and system-level dynamics of healthcare fraud:

- **The Fraud Triangle:** Evaluates pressure (financial strain on providers), opportunity (weak digital controls), and rationalization (perceived systemic inefficiencies).
- **The Theory of Planned Behavior (TPB):** Analyzes intent based on subjective norms and perceived behavioral control within the hospital management system.
- **The Technology Acceptance Model (TAM):** Measures how the adoption rate of anti-fraud systems by auditors is directly dependent on perceived ease of use and institutional utility.

## 3.2 Hypothesis Development

**H1:** States with high rates of Electronic Health Record (EHR) adoption under the Ayushman Bharat Digital Mission (ABDM) exhibit lower provider-level fraud rates.

**H2:** State-specific machine learning models optimize feature weights more accurately and achieve higher classification performance than generalized, all-India pooled models.

**H3:** Hybrid frameworks (combining supervised and unsupervised components) outperform purely supervised architectures in detecting novel, un-labeled fraud typologies.

**H4:** Distributed big data architectures achieve sub-second real-time risk scoring latencies without compromising classification accuracy metrics.

## 4. Research Methodology

### 4.1 Overall Design

This research adopts a mixed-methods explanatory sequential design. High-volume quantitative claim datasets are processed through predictive models to identify structural anomalies, followed by qualitative validation based on auditor feedback to contextualize the findings.

### 4.2 Data Sources & Sample Design

The quantitative dataset draws from multiple consolidated administrative registries and state portals:

| Data Source                                   | Primary Attributes Extracted                                                                                |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| National Health Authority (NHA)               | Transactional volumes, claim submission amounts, historical rejection rates, provider empanelment metadata. |
| Ministry of Health and Family Welfare (MoHFW) | Facility-level clinical infrastructure metrics, service utilization, bed capacities.                        |
| National Family Health Survey (NFHS-5)        | Regional health indicators, baseline insurance coverage ratios, household digital access indices.           |
| ABDM Health Stack Registries                  | Healthcare Professional Registry (HPR), Health Facility Registry (HFR), and facility EHR adoption tiers.    |
| RSBY Legacy Claims (2008–2018)                | Historical longitudinal baselines of public health insurance fraud patterns in India.                       |

The study implements a stratified random sampling methodology targeting 400,000 total claims (100,000 per selected state) across four distinct geographic regions reflecting diverse digital maturity tiers:

- **High Digital Maturity:** Tamil Nadu & Kerala (Robust digital history, high EHR saturation)
- **Moderate Digital Maturity:** Madhya Pradesh
- **Low Digital Maturity:** Uttar Pradesh

Additionally, a qualitative survey sample of N = 412 medical auditors and State Health Agency stakeholders was conducted to validate model features.

### 4.3 Variables and Feature Engineering

Independent Variables: State Digital Maturity Score, facility EHR adoption percentage, historical claim digitization rates, provider-to-bed ratios.

Dependent Variables: Binary fraud classification label (0 = Legitimate, 1 = Fraudulent), clinical claim-rejection rate, model performance metrics (AUC-ROC, Precision, Recall).

### 4.4 Machine Learning Configuration

The analytical pipeline was built to address severe class imbalance (fraudulent events comprising <1.5% of baseline records). Models include GBClassifier, Random Forest, Isolation Forest, and LSTM Anomaly Transformers using PySpark and Scikit-Learn.

## 5. Results and Discussion

### 5.1 Empirical Analysis of Digital Maturity

An evaluation of spatial records reveals a distinct negative correlation between digital health infrastructure investment and systemic fraud rates. States operating on centralized EHR standards show significantly lower baseline fraud metrics compared to regions characterized by fragmented, manual paper-trail validation.

### 5.2 Algorithmic Performance Benchmarks

| Model Architecture           | Precision | Recall | F1-Score | AUC-ROC |
|------------------------------|-----------|--------|----------|---------|
| Random Forest (Baseline)     | 0.88      | 0.84   | 0.86     | 0.92    |
| Gradient Boosted Trees (GBT) | 0.91      | 0.89   | 0.90     | 0.94    |
| LSTM Anomaly Transformer     | 0.93      | 0.95   | 0.94     | 0.96    |

## 6. Operational and Policy Implications

The National Health Authority (NHA) should embed the optimized LSTM model directly into the central TMS (Transaction Management System) to score claims during the pre-authorization phase. State Health Agencies can deploy analytical dashboards to route limited human auditing resources toward high-risk providers automatically.

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